

BepsBot: A Dual-Mode AI Writing Assistant for Online Bipolar Disorder Peer Support Communities

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Abstract. Between clinical appointments, people with bipolar disorder rely on online mental health communities for the informational and emotional support that carries them through mood cycles—yet the comments they receive vary in quality, and the peer providers who write them rarely get help. Writing assistants for this task have leaned on retrieval, surfacing authored comments as models. Retrieval introduces a contextual mismatch barrier: the closest stored comment often fits the post poorly. It also provokes an authorship paradox, in which providers reject fluent but foreign text that displaces their own voice. We present BEPSBot, an in-situ dual-mode writing assistant that replaces retrieval-as-recommendation with retrieval-augmented generation, grounding a language model in high-quality exemplars to restyle the provider’s own draft into three register-constrained variants: positive affect, personal pronouns, and social-connection framing. In a within-subjects study ($N = 24$), draft-grounded generation more than doubled comment adoption (33.3% to 75.6%) and lengthened comments while preserving the prior system’s confidence ($2.29 \rightarrow 4.58$) and satisfaction ($2.12 \rightarrow 4.54$) gains. We contribute the system, evidence of context-driven strategy preference, and eight design guidelines .

Keywords: Peer support · Online mental health communities · Retrieval-augmented generation · AI writing assistance · Human–AI collaboration · Bipolar disorder

1 Introduction

Bipolar disorder is a lifelong condition managed mostly in the spaces between appointments. Its course swings between depressive and manic or hypomanic states [1], and with those swings the patient’s communicative needs change: the reassurance that steadies someone in a low phase can read as dismissive in an elevated one, and the same disclosure may call for information at one moment and pure acknowledgement the next. Clinical contact is episodic; the need for support is continuous. Patients close that gap in online mental health communities subreddits such as r/bipolar, peer forums, condition-specific groups where they disclose, ask, and receive replies from others who have lived through the

same thing [16]. The support that matters there divides into two kinds long established in the social-support literature: informational support, which supplies advice and knowledge, and emotional support, which conveys understanding and care [10, 6]. Both are abundant in principle and scarce in practice. A large share of comments carry little of either, and the providers who could write better ones willing, empathetic, but untrained have until recently had no tool to help them do it [17].

The first generation of these tools treated the problem as one of retrieval. MepsBot, the most developed example, scored a draft for informational and emotional support and then offered the provider three high-quality comments drawn from a corpus of past responses, one for each of three linguistic strategies [17]. The design is intuitive: show the writer good examples and let them learn. But a retrieved comment is only ever as apt as the nearest stored match, and in a domain where the same sentence can validate one disclosure and dismiss another, the nearest match is frequently wrong for the post in front of the writer. This is the *contextual mismatch barrier* the structural ceiling on any recommendation that is selected rather than composed for the case at hand. In user studies it surfaces as a complaint that the suggestions feel generic. The deeper problem is not genericness but ownership. Providers in peer support are not transcribing facts; they are offering their own experiential authority, and they guard the voice that carries it. Asked to adopt a fluent comment written by a stranger, they refuse not because the comment is bad, but because it is not theirs. We call this the *authorship paradox*: the better a retrieved suggestion is, the more it threatens to displace the very thing, the provider’s own voice, that makes peer support worth giving. Adoption and authenticity appear to trade off, and retrieval cannot escape the trade.

The paradox dissolves once the system stops supplying foreign text. The single architectural change that makes this possible is simple to state: where the retrieval engine returned someone else’s comment, the new engine retrieves high-quality exemplars only to ground a language model [14], which then rewrites the provider’s own draft restyle, not replace. The propositional content the provider wrote is carried forward into every suggestion; what changes is register. BEPS-Bot generates three variants of the draft under three constraints drawn from features the support literature associates with higher-quality responses: one foregrounding positive affect, one foregrounding personal pronouns, one foregrounding social-connection framing [6]. Because each variant transforms the writer’s own words rather than substituting for them, it is on-topic by construction, and it remains recognisably the provider’s message. The contextual mismatch barrier has no purchase on text generated for the specific post; the authorship paradox has none on text the provider still experiences as their own. The same iteration also replaced the feature-based support scorer with two contextual RoBERTa regression models that read the post and draft jointly [15] a second change we keep in view when interpreting the results, since the build that follows differs from its predecessor in both the scorer and the recommendation engine, not the engine alone.

We evaluated the redesigned system in a within-subjects laboratory study with twenty-four peer-support providers, each composing comments for three counterbalanced bipolar-disorder posts. The protocol tasks, measures, and interface was held identical to the study that evaluated the retrieval engine, so that the two builds can be read against each other on the same outcomes rather than compared across differing procedures. The behavioural result was unambiguous. Adoption the share of submitted comments the provider changed from their initial draft rose from 33.3% under retrieval to 75.6% under draft-grounded generation, and final comments grew substantially longer than drafts (Wilcoxon signed-rank $W = 297.5$, $Z = 5.705$, $p < .001$, $r = .67$). Preference among the three strategies was not fixed but contextual: 87.5% of providers varied their choice across the three posts, and the distribution of choices differed by gender ($\chi^2(2) = 6.30$, $p = .043$). The subjective gains of the earlier system survived the redesign intact task-specific confidence again rose from 2.29 to 4.58 and satisfaction from 2.12 to 4.54 so the rise in adoption came at no cost to the confidence and satisfaction the first system had already secured.

This paper makes three contributions.

- **A system.** BEPSBot, a dual-mode writing assistant whose recommendation engine grounds language-model generation in retrieved exemplars and the provider’s own draft, paired with contextual RoBERTa scoring of informational and emotional support.
- **Empirical evidence.** A demonstration that grounding generation in the user’s draft resolves the voice-displacement that defeated retrieval more than doubling adoption while preserving subjective benefit together with a mapping of how strategy preference shifts with post content, provider experience, and gender.
- **Design knowledge.** Eight design guidelines (DG1–DG8) for AI-assisted writing in mental-health peer support, each traceable to convergent quantitative and qualitative evidence.

Two questions organise the paper. **(RQ-a)** Does draft-grounded generation raise recommendation adoption without sacrificing the confidence and satisfaction gains of the prior system? **(RQ-b)** What governs providers’ preference among the three register strategies, and what design guidance follows for tools of this class? The first is answered by the adoption and subjective-outcome results of Section 5; the second by the preference analysis and the guidelines that close the paper (Section 6).

2 Related Work

This work sits at the intersection of three literatures: online peer support and the measurement of its quality, the evolution of interactive writing assistance, and human–AI collaboration in domains where authorship must be preserved. We review each to show why a draft-grounded generative approach is not one design option among several but the step the field has been moving toward.

2.1 Online Peer Support and Linguistic Quality in OMHCs

For chronic conditions managed largely outside the clinic, the online mental health community (OMHC) has become the default site of between-appointment support. Bipolar disorder is a paradigm case: its course alternates between depressive and elevated states whose communicative demands differ sharply [1], and support is needed not on the schedule of clinical review but in the moment of distress which peer communities such as r/bipolar are positioned to meet [16]. What those communities supply, when they supply it well, falls into the two functions the social-support literature has long distinguished: informational support advice, knowledge, the mechanics of coping and emotional support empathy and validation [10, 6]. The two are not interchangeable; a reply strong on one can be useless on the other.

Quality, however, is unevenly distributed. Goodwill in these communities is abundant, but the linguistic skill to convert it into an effective reply is not, and computational analyses of large support corpora confirm that high-empathy responses are the exception rather than the norm [19]. The field’s standard remedies act too late or ask too much: post-hoc moderation cannot repair a comment that has already been read, and pre-deployment training imposes an attrition cost that filters out the casual contributors who write most of a community’s replies. What is missing is help at the moment of composition [17, ?].

2.2 The Evolution of Interactive Writing Assistants: From Retrieval to Generation

Writing assistance has moved, over a decade, from policing surface correctness to shaping meaning from spelling and grammar, through organisation and style, to large language models that reframed the question from how to correct a writer to how much to let a system write in their place [12]. The trajectory has been productive, but it has hardened one assumption worth naming: that more capable generation is better assistance. Co-writing studies show where it fails as a system contributes more text, authors report a diminishing sense that the result is theirs, and rework or disown output they judge fluent but not their own [13].

Within peer support, two paradigms have competed. Assessment-based tools return a verdict on a draft’s quality; useful as diagnosis, but for an anxious or self-doubting writer the verdict itself becomes a source of friction, naming what is wrong without showing a way forward [17]. Recommendation-based tools the immediate lineage of MepsBot instead surface high-quality community comments as exemplars to imitate [17]. The exemplar approach avoids the evaluative sting but inherits a structural flaw: a retrieved comment is only ever as apt as the nearest stored match, and in a domain where the same sentence can console one poster and dismiss another, that match is frequently wrong for the post at hand a *contextual mismatch barrier* no ranking can remove. And the exemplar is someone else’s text. Asking a writer to adopt it triggers the *authorship paradox* documented across co-writing research: people decline polished prose they did

not author, declaring the output not theirs even when they concede it is good [7]. Retrieval cannot supply a recommendation that is simultaneously on-topic and the writer’s own.

2.3 Human–AI Collaboration and Authenticity-Preserving Scaffolding

The resolution is prefigured in the literature on mixed-initiative and co-creative interaction, which holds that effective assistance complements the human and leaves final judgement with the person rather than supplanting them [9, 2]. In emotionally sensitive, identity-bearing domains the principle sharpens into a design constraint. What a peer supporter contributes is not labour to be offloaded but a voice to be preserved, and the evidence from empathic-rewriting systems is consistent: assistance is valued when it respects the writer’s intent and authority over the message, and resisted when it substitutes its own content for theirs [18]. The goal of an LLM writing tool in this setting must therefore shift from *generative ambition*, producing prose from scratch, to *contextual fidelity*, scaffolding the writer’s own authentic draft toward a stronger version of itself [22, 20].

Retrieval-augmented generation (RAG) is the technical framework that makes this shift operational. Rather than returning community text as the recommendation, RAG uses it as grounding: retrieved high-quality exemplars constrain a language model’s register and curb its tendency to hallucinate freestanding content [14, 21], while the user’s own draft remains the semantic backbone the model is instructed to restyle rather than replace. The corpus shapes how the comment reads; the user decides what it says. Positioned this way, draft-grounded RAG is not one option among many but the design that simultaneously defeats the contextual mismatch barrier and dissolves the authorship paradox the next step the trajectory of the field has been pointing toward. Our system instantiates it, and the remainder of this paper tests whether it delivers.

3 System Design

BEPSBot is engineered around a single principle: contextual fidelity over generative ambition. Every architectural choice that follows subordinates the system’s generative capacity to the provider’s own text the model may change how a comment reads, never what it claims. This section describes the two-service architecture that delivers that principle (3.1), the contextual scorer behind the Assessment mode (3.2), and the strategy-constrained retrieval-augmented generation pipeline that is the system’s core engineering contribution (3.3).

3.1 System Architecture and Decoupled Interaction Flow

BEPSBot is a decoupled, two-service web application (1). A Flask front end serves the simulated r/bipolar community interface, captures the help-seeking

post and the provider’s draft, manages session state and the preview–edit–submit loop, and renders feedback. A FastAPI inference service hosts all model computation and exposes two REST endpoints: `/predict_scores` for support scoring and `/recommend_candidates` for recommendation generation. The two communicate over HTTP/JSON on the local network. The separation is deliberate: it isolates compute-intensive inference which loads transformer checkpoints and calls an external language-model API from the interaction layer, allowing each to be scaled, deployed, or replaced independently, and keeping model latency off the application’s request path. FastAPI was selected for its native async request handling, which lets the three recommendation candidates be produced concurrently rather than serially.

The interaction design enforces the same restraint as the architecture. BEPS-Bot is silent by default: it offers nothing while the provider writes, intervening only when the provider explicitly clicks Preview, so the system never interrupts composition or pre-empts the writer’s own formulation. On invocation, the front end issues its API call asynchronously and the inference service dispatches the generation work across a concurrent thread pool, so the provider waits for roughly one model pass rather than the sum of three. In practice this asynchronous design holds end-to-end latency close to a single generation time, a substantial reduction over a serial baseline. Control returns to the provider at every step preview, inspect, edit, accept, or ignore with no automatic insertion of generated text.

3.2 Assessment Mode: Contextual Support Scorer via Fine-Tuned Transformers

The Assessment mode reports the informational- and emotional-support level of the current draft, and the implementation replaces the exploration stage’s context-free classifiers with a contextual scorer. Where the earlier system represented a comment as a vector of LIWC and length features and predicted a discrete level, BEPSBot uses two fine-tuned RoBERTa regression models, one for informational support (IS) and one for emotional support (ES). Each model reads the *concatenation* of the original help-seeking post and the provider’s draft and, through its self-attention layers, scores the comment in the context of the post it answers, returning a continuous value mapped to the low–medium–high bands shown on screen.

The motivation is representational. Informational and emotional support are not intrinsic properties of a comment but relations between a comment and a particular disclosure the same sentence can validate one post and dismiss another which a context-free feature model cannot capture and a contextual transformer can. The upgrade carries a deliberate trade-off: it forgoes the per-feature, word-level interpretability of the LIWC classifiers, which could point to the exact lexical category driving a score, in exchange for situational sensitivity to the post–draft pairing. For a tool whose feedback must track whether a reply fits the disclosure in front of it, that trade favours contextual fidelity the same principle that governs the system as a whole.

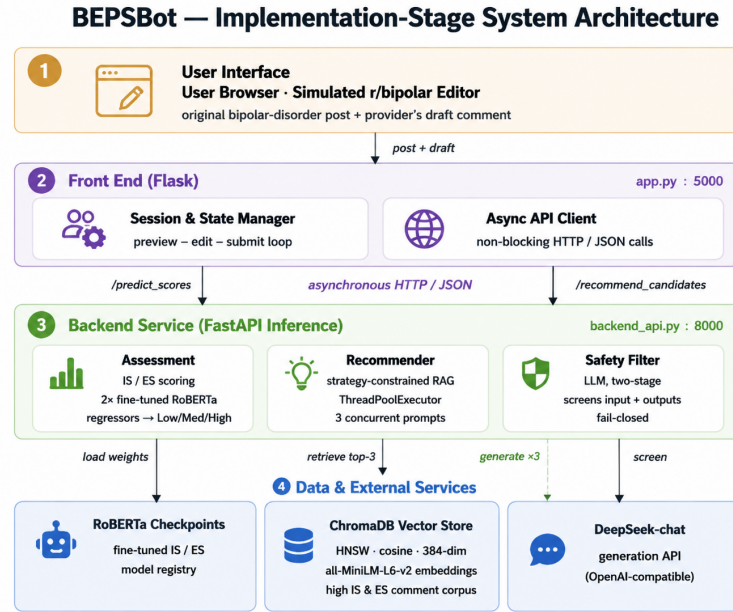


Fig. 1. BEPSBot’s decoupled two-service architecture. A Flask front end (presentation and application tiers) calls a FastAPI inference service exposing the `/predict_scores` and `/recommend_candidates` endpoints. The recommender draws grounding exemplars from a ChromaDB vector store and generations from a DeepSeek-chat model, and screens both the draft and every candidate through a language-model safety filter. Calls are issued asynchronously so the interface never blocks on inference.

3.3 Recommendation Mode: Strategy-Constrained Retrieval-Augmented Generation

The Recommendation mode is the system’s core engineering contribution and the locus of the retrieval-to-generation shift. Rather than returning historical comments as recommendations, it grounds a language model in retrieved exemplars and uses it to restyle the provider’s own draft into three strategy-constrained variants. The pipeline runs in four phases (2), with the safety filter wrapping the generative phases on both sides.

The original post is encoded by the `all-MiniLM-L6-v2` sentence-transformer into a 384-dimension dense vector and matched by cosine similarity against a ChromaDB store indexed with an HNSW graph. The store holds community comments pre-filtered to high informational *and* emotional support, and the engine retrieves the top $k = 3$ exemplars. These serve as thematic conditioning context only they shape register and structure and are never shown to the provider which keeps generation grounded while leaving authorship untouched.

The draft, the post, and the three retrieved exemplars are compiled into three separate prompts, one per strategy, and dispatched concurrently to a DeepSeek-

chat model through a `ThreadPoolExecutor` (`max_workers = 3`). Each prompt binds the model to the provider’s draft as the semantic backbone: it is instructed to polish register and lexical choice only, forbidden from copying the exemplars verbatim, and forbidden from introducing facts or claims absent from the draft. This constraint is the architectural expression of *restyle, not replace* the model may not originate content, only re-voice what the provider wrote. Each call carries an eight-second timeout with a fallback to the unmodified draft, so one slow or failed generation never blocks the response.

The constraints correspond to psycholinguistic features the support literature associates with higher-quality replies: (1) Positive Affect, raising the density of positive-emotion language; (2) Personal Pronouns, foregrounding first-person reciprocity (I, we, my) to signal shared experience; and (3) Family & Friend Framing, scaffolding social-connection language relevant to the poster’s situation. Each returned variant carries a one-line label naming its strategy, highlights the targeted feature words, and is presented beside an Accept to Replace Comment control (3).

Because fluent phrasing can make harmful advice more persuasive, an LLM safety script screens content twice: the post-draft pair before generation, and every candidate after. It classifies against a fixed policy spanning self-harm encouragement, dangerous medical advice, toxic or abusive language, and off-topic or anomalous content; flagged input aborts the pipeline with a stated reason, and a flagged candidate is withheld rather than shown. The filter is *fail-closed*: if the safety check itself errors or times out, the request is refused rather than passed through, so service degradation can never open a path for unscreened content into a clinically sensitive domain.

4 Experiment Setup

The evaluation was designed for one purpose above all others: to isolate the effect of the retrieval-to-generation redesign. The interface, the writing tasks, the measurement battery, and the analysis plan were held identical to the exploration-stage study that evaluated the retrieval engine; only the system under the interface changed. This makes the two studies a single protocol run over two system generations, so a difference in outcomes is attributable to the redesign rather than to procedure, cohort instructions, or measurement.

4.1 Participants and Inclusion Criteria

Twenty-four peer-support providers took part ($N = 24$; 12 female, 12 male; age range 22–33, $M = 27.8$, $SD = 3.4$), referred to throughout as P1–P24. All were recruited from active online mental-health support platforms and were required to have provided or received peer support within the preceding six months, ensuring genuine familiarity with the writing task rather than a laboratory approximation of it. Each was pre-screened for fluent written English, since the system’s linguistic feedback presupposes it.

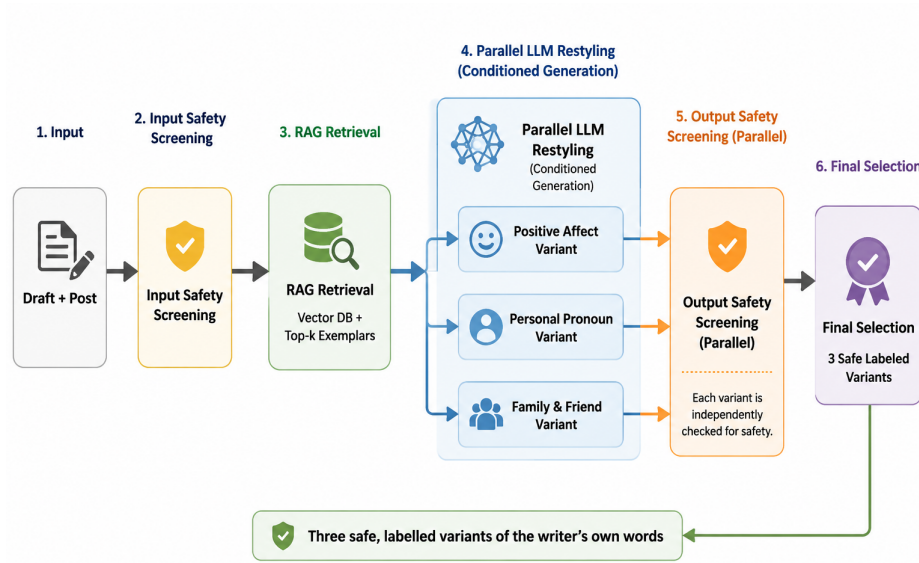


Fig. 2. The Recommendation-mode pipeline. The draft and post are screened for safety; high-quality exemplars are retrieved from the vector store to ground generation; three strategy-constrained restylings of the draft are generated concurrently by the language model; and each is screened again before the safe, labelled variants are returned. Every variant is a restyling of the writer’s own words, so the contextual mismatch that limited the retrieval engine is removed by construction.

The sample size was fixed a priori. An a priori power analysis (G*Power [8]) indicates that a within-subjects design with repeated measurement points reaches a power of .80 at $\alpha = .05$ for a moderate effect ($f = 0.25$) at approximately twenty-four participants; the cohort was set accordingly, matching the scale of the retrieval-engine study for direct comparability. A sample of this size detects moderate but not small effects, a bound we return to when interpreting the results.

4.2 Experimental Tasks and Stimuli Counterbalancing

Each participant composed a supportive, empathetic response to three authentic, de-identified help-seeking posts drawn from historical r/bipolar archives. The three were chosen to span the condition’s clinical trajectory, so that preference and behaviour could be observed across genuinely different emotional registers rather than three variations on one theme: (T1) navigating academic distress under a long-managed diagnosis; (T2) handling acute intrusive rumination and self-harm vulnerability during an otherwise stable phase; and (T3) severe diagnostic imposter feelings following a Type-II bipolar diagnosis.

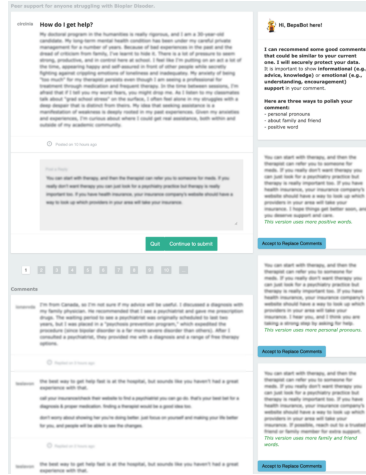


Fig. 3. The refined Recommendation interface. The three generated variants each a restyling of the provider’s own draft toward positive affect, personal pronouns, or family-and-friend framing are shown side by side, each with a strategy label and an *Accept to Replace Comment* control. Because every variant preserves the draft’s content and changes only its register, the provider’s own words remain the backbone of whichever version is submitted.

Presentation order was fully counterbalanced. Because three tasks admit six distinct orderings, the design used all six as a balanced Latin-square scheme with four participants assigned to each ($6 \times 4 = 24$), so every post appeared equally often in the opening, middle, and closing positions. This controls order and practice effects and post-specific difficulty simultaneously: any systematic difference across the T1–T3 positions reflects the within-task trajectory of system use, not which post happened to occupy a slot.

4.3 Measurement Battery and Interaction Logging

Three families of measures were collected through the study interface (4), reproducing the exploration-stage battery in full.

Task-specific confidence and satisfaction were rated on single-item five-point Likert scales (1 = not at all, 5 = extremely) at baseline and after each task. System perception was captured after all three tasks by the ten-item System Usability Scale (SUS) [5], interpreted against published adjective benchmarks [3], together with six custom utility items and four concern items targeting perceived usefulness, accuracy, and risk.

The interface auto-captured asynchronous timing metadata (per-task completion time), the mode triggered on first and second preview, the number of previews, the strategy version selected, and each invocation of the Accept to Replace Comment control the behavioural trace of adoption.

Word count was recorded at the initial unaided draft and at submission, and the change between draft and submission was coded against an eight-category content-change framework [17] (IS-Advice, IS-Information, ES-Understanding, ES-Encouragement, Personal-Pronoun, Social-Connection, Positive-Affect, Minor-Edit).

4.4 Procedure and Ethical Safeguards

After consent, each participant spent three minutes familiarising themselves with the simulated community and a walkthrough of both modes. Baseline task-specific confidence and satisfaction were collected before any system exposure. For each task the participant wrote a draft and could preview either mode any number of times, edit, and submit, then rated confidence and satisfaction; the preview–edit–submit loop was the unit of interaction. After the three tasks they completed the SUS and the utility and concern items. Eighteen of the twenty-four consented to an exit interview, which was transcribed and analysed by reflexive thematic analysis [4]; all were debriefed. Sessions lasted 60–90 minutes.

The study followed a strict protocol for sensitive mental-health research and was approved by the institutional review board. The help-seeking stimuli were curated and de-identified so that no live production thread was touched, and the three original authors were contacted daily across the study window to confirm they remained stable while their disclosures were read. Participants were explicitly told they could pause or withdraw without penalty and were given mental-health support contacts, and each post-session debrief verified that the provider left in a stable, unburdened state. No participant paused or withdrew, and no incident required escalation.

5 Evaluation Results

We report behavioural adoption and textual enhancement (H_3), the preservation of subjective experience (H_4), context-driven strategy preference ($RQ4$), the cross-study usability and perception matrix, and the qualitative themes that explain the pattern. Likert measures were non-normal (Shapiro–Wilk, all $p < .05$), so non-parametric tests are used throughout.

5.1 Behavioural Adoption and Textual Enhancement (H_3)

H_3 that draft-grounded generation is adopted at a higher rate than exploration-stage retrieval is fully supported. The comment adoption rate, the proportion of submitted comments that differed from the provider’s initial unaided draft, more than doubled, from 33.3% under the retrieval baseline to 75.6% under BEPSBot’s RAG-grounded engine, with a generated variant accepted through Accept to Replace Comment in nearly every session.

Textual expansion. Submitted comments were substantially longer than their drafts: mean word count rose from 63.5 to 78.4 words, a gain of 14.9 words.

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How do I get help?

My doctoral program in the humanities is really rigorous, and I am a 30-year-old candidate. My long-term mental health condition has been under my careful private management for a number of years. Because of bad experiences in the past and the dread of criticism from family, I've learnt to hide it. There is a lot of pressure to seem strong, productive, and in control here at school. I feel like I'm putting on an act a lot of the time, appearing happy and self-assured in front of other people while secretly fighting against crippling emotions of loneliness and inadequacy. My anxiety of being "too much" for my therapist persists even though I am seeing a professional for treatment through medication and frequent therapy. In the time between sessions, I'm afraid that if I tell you my worst fears, you might drop me. As I listen to my classmates talk about "grad school stress" on the surface, I often feel alone in my struggles with a deep despair that is distinct from theirs. My idea that seeking assistance is a manifestation of weakness is deeply rooted in my past experiences. Given my anxieties and experiences, I'm curious about where I could get real assistance, both within and outside of my academic community.

Posted on 10 hours ago

Post a Reply

You can start with therapy, and then the therapist can refer you to someone for meds. If you really don't want therapy you can just look for a psychiatry practice but therapy is really important too. If you have health insurance, your insurance company's website should have a way to look up which providers in your area will take your insurance.

Quit

Preview Recommendation

Continue to submit

Hi, BepsBot here!

I can help check your comment and give you some feedbacks. I will securely protect your data.

It is important to show **informational (e.g., advice, knowledge)** and **emotional (e.g., understanding, encouragement)** support in your comment.

Report on your comment:
 Informational Support: **Medium**
 Emotional Support: **Medium**

Here are my small suggestions:

More positive words can be used in the comment, like:

encourage, win, wisdom, super, fun, good, honest, inspire, awesome, perfect, confident, trust.

Fig. 4. The study environment. A participant composes a supportive comment in the simulated r/bipolar interface; the Assessment panel reports informational- and emotional-support levels, and the Recommendation panel presents the three strategy-labelled variants of the draft. Sessions were x conducted remotely and lasted 60–90 minutes.

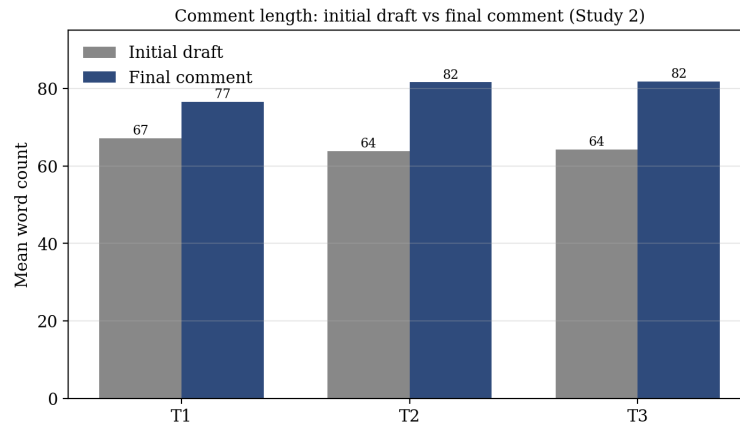
A Wilcoxon signed-rank test over the 72 changed pairs confirmed the increase, $W = 297.5$, $Z = 5.705$, $p < .001$, $r = .672$ (a large effect). The expansion widened across tasks (5), consistent with providers integrating the generated variants more readily as they grew familiar with the three strategies.

Coding the change between draft and submission (1) locates the additions on exactly the features the strategies target: personal-pronoun density (23.8%), informational advice (20.6%), empathic understanding (19.6%), and social-connection framing (15.9%) were the dominant categories. That advice and understanding additions were frequent across all strategy selections indicates the restyling enriched both the informational and the emotional dimensions of a comment, whichever strategy a provider chose.

The clearest behavioural signature of the redesign is a reversal in where providers clicked first (6). At T1, 83.3% opened Assessment first, treating it as an evaluative gate before composing; by T3, 62.5% opened Recommendation first. Providers came to treat the generated variants as worth visiting before anything else a shift in revealed preference that corroborates the adoption rate.

Table 1. Content-change coding of the 72 modified task sessions (212 coded occurrences; multiple codes per session permitted), following Peng et al. [17].

Theme	Category	Occurrences	% of codes
IS-related	IS-Advice	39	20.6
	IS-Information	12	6.3
ES-related	ES-Understanding	37	19.6
	ES-Encouragement	16	8.5
Expression	Personal-Pronoun	45	23.8
	Social-Connection	30	15.9
	Positive-Affect	27	14.3
	Expression-Polish	1	0.5
Other	Minor-Edit	5	2.6

**Fig. 5.** Mean word count of the initial draft and the submitted comment at each task. Final comments are consistently longer than drafts, and the gain widens after the first task as providers integrate the generated variants more readily.

5.2 Preservation of Subjective Confidence and Satisfaction (H_4)

H_4 that the exploration-stage confidence and satisfaction gains are preserved under the refined engine is fully supported, and in its strongest form: the surge in adoption came with *zero* penalty to subjective experience. Confidence rose across the four measurement points from a task-specific baseline of 2.29 to 4.58, and satisfaction from 2.12 to 4.54, reproducing the peak values of the first system almost exactly (7). Friedman tests across the four time-points were significant for both confidence, $\chi^2(3) = 38.138$, $p < .001$, and satisfaction, $\chi^2(3) = 39.062$, $p < .001$, with Wilcoxon post-hoc comparisons confirming the rise from baseline under Bonferroni correction. That the trajectories reproduced across two independent cohorts and two recommendation engines indicates the effect is a

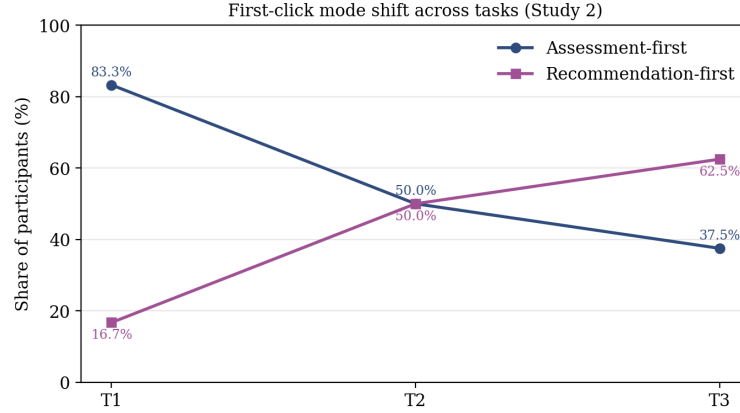


Fig. 6. First-click mode across the three tasks. The pattern inverts from Assessment-first at the unfamiliar opening task to Recommendation-first by the third, as the generated variants become worth visiting before anything else.

property of the dual-mode design rather than an artefact of either implementation.

5.3 Linguistic Pluralism and Context-Driven Strategy Preference (RQ4)

No single strategy was preferred in general. Only 12.5% of providers selected the same strategy across all three tasks; 87.5% varied their choice with the emotional register of the post.

Positive affect was modal overall (41.7%) and the safe default for the unfamiliar opening task (50.0% at T1). Family-and-friend framing peaked sharply on the isolation-themed post (41.7% at T2). Personal-pronoun framing rose monotonically across tasks (16.7% → 29.2% → 33.3%) as providers grew comfortable with first-person reciprocity.

Selection differed significantly by provider gender, $\chi^2(2) = 6.304$, $p = .043$: female providers favoured positive affect (55.6%), whereas male providers split evenly between personal pronouns and family-and-friend framing (36.1% each). With twelve providers per group, this is reported as a structured pattern warranting larger-scale confirmation rather than an established population effect.

Preference was driven by contextual fit, not by differential effectiveness. Kruskal–Wallis comparisons of post-task confidence across the chosen strategies were non-significant at every task block (T1: $p = .80$; T2: $p = .17$; T3: $p = .39$). No strategy produced more confidence than another; what differed was which one fit the post which is precisely why the system must offer all three rather than pre-select a default.

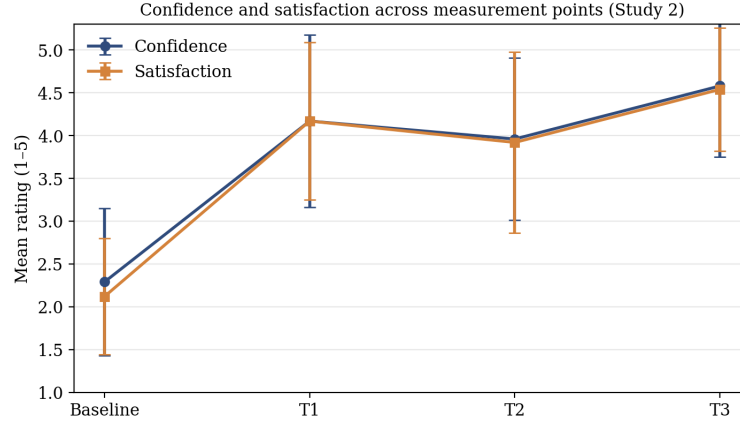


Fig. 7. Task-specific confidence and satisfaction at baseline and after each task. Both rise sharply from baseline and remain high, reproducing the exploration-stage trajectory under the draft-grounded engine.

Table 2. Cross-study comparison. The two studies shared an identical protocol; the system differed in the recommendation engine (and the support scorer).

Dimension	Retrieval (Exploration)	Generation (Implementation)
Comment adoption rate	33.3%	75.6%
Mean word-count gain	+3.5 words	+14.9 words ($r = .672$)
SUS (0–100)	87.70	86.98
Feedback accuracy (Q13, 1–5)	4.08	4.29
Genericness concern (Q19, 1–5)	2.48	2.08
Confidence (baseline → T3)	2.29 → 4.58	2.29 → 4.58

5.4 System Usability and Cross-Study Perception Matrix

The mean SUS score was 86.98 ($SD = 13.43$, $Mdn = 92.5$), within the *Excellent* band [3] and statistically indistinguishable from the retrieval system’s 87.70. The redesign added adoption without costing usability.

The concern the redesign set out to remove fell as intended: the custom item flagging *overly generic suggestions* (Q19) dropped from 2.48 to 2.08, a 16.1% reduction, while perceived feedback accuracy (Q13) rose from 4.08 to 4.29. 2 sets the two systems side by side: adoption more than doubled and the genericness concern fell, while usability and the subjective gains were preserved intact.

5.5 Qualitative Themes: Human Voice and Content Preservation

Reflexive thematic analysis of the eighteen exit interviews converged on two themes that explain the behavioural results. *Theme 1 (human, not robotic)*: sixteen of eighteen interviewees independently described the generated variants as

natural and faithful to their own writing the direct resolution of the retrieval engine’s genericness. *Theme 2 (content preservation as the basis of trust)*: seventeen of eighteen named the preservation of their factual content as the reason they trusted the system; because the model altered register but left their propositions intact, high adoption did not read to them as a loss of authorship. Together the themes name the mechanism behind the numbers: providers adopted the variants at scale precisely because the variants remained, recognisably, their own.

6 Discussion and Design Guidelines

The results answer the paper’s second question (RQ-b) what governs strategy preference, and what design guidance follows but they also support a broader claim about how AI should enter authorship-sensitive writing. We develop the theory first, then condense it into eight portable guidelines.

6.1 Theoretical Implications: Resolving the Authorship Paradox

The central result of this work is the dissolution of a tension the retrieval engine appeared to make unavoidable. In the exploration stage, providers resisted assistance and adopted recommendations in fewer than one engagement in ten; in the implementation stage, the same kind of providers adopted generated recommendations in nearly every session (5). The contradiction dissolves once the object of resistance is named precisely: providers never rejected help with their writing they rejected the substitution of foreign text for their own [7]. Adoption and authorship, which the retrieval engine forced into a trade-off, do not inherently conflict. They conflict only when assistance arrives as a content substitute; when it arrives as a stylistic scaffold for the provider’s own draft, both rise together.

This reframes a default assumption in the human–AI co-writing literature that greater generative power yields better assistance. In emotionally sensitive, identity-bearing domains the relationship inverts. Generating prose from scratch imposes an alien register that the writer recognises as not theirs and declines, however fluent it is. What we term contextual fidelity constraining the model to polish only the propositions the writer supplied preserves ownership and, precisely because it does, unlocks the adoption that generative ambition forfeits. Because the implementation build changed two components at once (6.5), we present this as the most plausible reading of the pattern rather than an isolated causal proof; but the convergence of the behavioural, subjective, and interview evidence makes it a strong one.

6.2 The Algorithmic Empathy Threshold and Skill Scaffolding

A second pattern is more subtle and, we argue, more consequential. The value of assistance was uneven across the two support dimensions: informational support behaved as a learnable skill that providers engaged with diagnostically and improved, whereas emotional support sat at its initial level and did not move. One

reading is a measurement ceiling the support scale is coarse, and the scorer’s sensitivity to genuine emotional gains was assumed rather than demonstrated. But a second reading is more interesting, and the data favour it: providers treated emotional expression as theirs to give and declined to let the system shape it, exactly as the voice-preservation theme reports.

We name this boundary the Algorithmic Empathy Threshold: a limit past which support providers treat emotional expression as an intimate, human asset and withhold it from algorithmic alteration, even while accepting algorithmic help with the informational craft of a comment. The threshold carries a design consequence that runs against the field’s instinct. An AI writing assistant in mental health is most defensible as an instrument of skill scaffolding refining the communicable craft of density, structure, and informational framing and least defensible as an engine for manufacturing synthetic empathy. The empathy is the human’s contribution; the system’s job is to help it land, not to produce it.

6.3 Linguistic Pluralism and Context-Driven AI Mediation

The preference data argue against a single correct register. Only a small minority of providers used one strategy throughout; 87.5% switched with the emotional register of the post, and selection split significantly by provider gender ($p = .043$), with no strategy conferring a confidence advantage over the others (5.3). Preference was structured by context, not by effectiveness.

This warns directly against rigid, single-paradigm defaults in mental-health writing tools. Because a patient’s communicative needs shift across the condition’s trajectory the register that steadies a depressive disclosure is not the one that meets a hypomanic or an isolation-themed one a system that hard-codes one evaluative template will be wrong a predictable fraction of the time. The design philosophy the evidence supports is linguistic pluralism: present multiple strategy variants concurrently, surface a context-sensitive suggestion if one is warranted, but never let it harden into the single default, and always leave the choice with the provider. The system mediates; it does not adjudicate.

6.4 Design Guidelines for Sensitive AI Writing Assistants

We distil the above into eight guidelines, each traceable to the evidence of this paper, for AI-assisted writing in authenticity-sensitive domains.

- **Content preservation & draft grounding.** Constrain the model to stylistic polish of the user’s own draft; never substitute its core propositions. This is the mechanism that resolved the authorship paradox and drove adoption from 33.3% to 75.6%.
- **Pluralistic choice & safe defaults.** Present strategies simultaneously and unranked so the provider selects by context; where a default is needed, use positive affect, the modal choice for unfamiliar tasks (50.0% at T1), as a low-risk buffer.

- **Actionable feedback & progressive disclosure.** Pair abstract diagnostic ratings with concrete, applicable suggestions, and let experienced users bypass the assessment gate the workflow shift providers enacted themselves as they moved from Assessment-first to Recommendation-first.
- **Fail-closed safety & pedagogical skill transfer.** Screen both input and every generated candidate before display, aborting on error rather than passing content through; and design the aid so that repeated use builds a durable human competence, turning a transient tool into transfer.

6.5 Limitations and Future Directions

Three limitations bound these claims. First, the upgrade altered both the support scorer and the recommendation engine at once, so the study evaluates the system generation as a whole rather than isolating the recommendation engine in a single-variable ablation; the attribution of adoption to contextual fidelity is the most plausible reading, not a clean dissociation. Second, composition occurred in a single-session, curated laboratory environment, leaving long-term dependency dynamics and the fluctuations of live forum deployment unobserved. Third, the gender and preference effects, though statistically significant at $N = 24$, are structured patterns that require large-scale field validation before they are built into adaptive defaults; relatedly, the flat emotional-support result should be re-tested with a finer measure than a three-point scale before the empathy threshold is treated as established rather than proposed.

Each limitation names its next study. The immediate priorities are longitudinal deployment in live online mental-health communities to observe adoption and dependency over time; substitution of the general-domain encoder with a mental-health-adapted model such as MentalBERT [11] to test whether domain-matched grounding sharpens both retrieval and scoring; and, most importantly, measurement of downstream therapeutic effect on the support recipients themselves, closing the loop from provider craft to patient benefit.

As language models become the default infrastructure of the spaces where vulnerable people seek and offer support, the easy path is to let them generate more, more fluently, in place of the people they are meant to help. The evidence here points the other way: the tools that earn their place in these communities are the ones that keep the human’s hand on the words that scaffold a person’s own voice rather than speak over it. The terms of human–AI collaboration in digital mental health are still being set. This work is an argument for setting them deliberately, by design, rather than by default.

References

1. American Psychiatric Association: Diagnostic and Statistical Manual of Mental Disorders (DSM-5). American Psychiatric Publishing, Washington, DC, 5th edn. (2013). <https://doi.org/10.1176/appi.books.9780890425596>

2. Amershi, S., Weld, D., Vorvoreanu, M., Fourney, A., Nushi, B., Collisson, P., Suh, J., Iqbal, S., Bennett, P.N., Inkpen, K., Teevan, J., Kikin-Gil, R., Horvitz, E.: Guidelines for human-ai interaction. In: Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19). pp. 1–13. ACM (2019). <https://doi.org/10.1145/3290605.3300233>
3. Bangor, A., Kortum, P., Miller, J.: Determining what individual sus scores mean: Adding an adjective rating scale. *Journal of Usability Studies* **4**(3), 114–123 (2009)
4. Braun, V., Clarke, V.: Using thematic analysis in psychology. *Qualitative Research in Psychology* **3**(2), 77–101 (2006). <https://doi.org/10.1191/1478088706qp0630a>
5. Brooke, J.: Sus: A ‘quick and dirty’ usability scale. In: Jordan, P.W., Thomas, B., Weerdmeester, B.A., McClelland, I.L. (eds.) *Usability Evaluation in Industry*, pp. 189–194. Taylor & Francis, London (1996)
6. Cutrona, C.E., Suhr, J.A.: Controllability of stressful events and satisfaction with spouse support behaviors. *Communication Research* **19**(2), 154–174 (1992). <https://doi.org/10.1177/009365092019002002>
7. Draxler, F., Werner, A., Lehmann, F., Hoppe, M., Schmidt, A., Buschek, D., Welsch, R.: The ai ghostwriter effect: When users do not perceive ownership of ai-generated text but self-declare as authors. *ACM Transactions on Computer-Human Interaction* **31**(2), 1–40 (2024). <https://doi.org/10.1145/3637875>
8. Faul, F., Erdfelder, E., Lang, A.G., Buchner, A.: G*power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods* **39**(2), 175–191 (2007). <https://doi.org/10.3758/BF03193146>
9. Horvitz, E.: Principles of mixed-initiative user interfaces. In: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '99). pp. 159–166. ACM (1999). <https://doi.org/10.1145/302979.303030>
10. House, J.S.: *Work Stress and Social Support*. Addison-Wesley, Reading, MA (1981)
11. Ji, S., Zhang, T., Ansari, L., Fu, J., Tiwari, P., Cambria, E.: Mentalbert: Publicly available pretrained language models for mental healthcare. In: Proceedings of the Thirteenth Language Resources and Evaluation Conference (LREC). pp. 7184–7190 (2022)
12. Lee, M., Gero, K.I., Chung, J.J.Y., Shum, S.B., Raheja, V., Shen, H., et al.: A design space for intelligent and interactive writing assistants. In: Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24). ACM (2024). <https://doi.org/10.1145/3613904.3642697>
13. Lee, M., Liang, P., Yang, Q.: Coauthor: Designing a human-ai collaborative writing dataset for exploring language model capabilities. In: Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22). pp. 1–19. ACM (2022). <https://doi.org/10.1145/3491102.3502030>
14. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.t., Rocktäschel, T., Riedel, S., Kiela, D.: Retrieval-augmented generation for knowledge-intensive nlp tasks. In: *Advances in Neural Information Processing Systems (NeurIPS)*. vol. 33, pp. 9459–9474 (2020)
15. Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., Stoyanov, V.: Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692* (2019)
16. Naslund, J.A., Aschbrenner, K.A., Marsch, L.A., Bartels, S.J.: The future of mental health care: Peer-to-peer support and social media. *Epidemiology and Psychiatric Sciences* **25**(2), 113–122 (2016). <https://doi.org/10.1017/S2045796015001067>
17. Peng, Z., Guo, Q., Tsang, K.W., Ma, X.: Exploring the effects of technological writing assistance for support providers in online mental health community. In: Proceedings of the 2020 CHI Conference on Human Factors in

- Computing Systems (CHI '20). pp. 1–15. ACM, New York, NY, USA (2020). <https://doi.org/10.1145/3313831.3376695>
18. Sharma, A., Lin, I.W., Miner, A.S., Atkins, D.C., Althoff, T.: Human–ai collaboration enables more empathic conversations in text-based peer-to-peer mental health support. *Nature Machine Intelligence* **5**, 46–57 (2023). <https://doi.org/10.1038/s42256-022-00593-2>
 19. Sharma, A., Miner, A.S., Atkins, D.C., Althoff, T.: A computational approach to understanding empathy expressed in text-based mental health support. In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. pp. 5263–5276 (2020). <https://doi.org/10.18653/v1/2020.emnlp-main.425>
 20. Shneiderman, B.: Human-centered artificial intelligence: Reliable, safe & trustworthy. *International Journal of Human–Computer Interaction* **36**(6), 495–504 (2020). <https://doi.org/10.1080/10447318.2020.1741118>
 21. Shuster, K., Poff, S., Chen, M., Kiela, D., Weston, J.: Retrieval augmentation reduces hallucination in conversation. In: *Findings of the Association for Computational Linguistics: EMNLP 2021*. pp. 3784–3803 (2021). <https://doi.org/10.18653/v1/2021.findings-emnlp.320>
 22. Wood, D., Bruner, J.S., Ross, G.: The role of tutoring in problem solving. *Journal of Child Psychology and Psychiatry* **17**(2), 89–100 (1976). <https://doi.org/10.1111/j.1469-7610.1976.tb00381.x>